

whose Fourier series has the partial sums

$$f_{2n-1}(x) = \frac{4}{\pi} \left[\sin x + \frac{1}{3} \sin 3x + \cdots + \frac{\sin(2n-1)x}{(2n-1)} \right].$$

To verify this for $f(x)$, proceed as follows:

(a) Show that

$$\begin{aligned} \pi(\sin x)f'_{2n-1}(x) &= 4 \sin x [\cos x + \cos 3x \\ &\quad + \cdots + \cos(2n-1)x] \\ &= 2 \sin 2nx. \end{aligned}$$

(b) Infer from part (a) and the figure that the maximum occurs at $x = \pi/(2n)$ and has the value

$$f_{2n-1}\left(\frac{\pi}{2n}\right) = \frac{4}{\pi} \left[\sin \frac{\pi}{2n} + \frac{1}{3} \sin \frac{3\pi}{2n} + \cdots + \frac{1}{2n-1} \sin \frac{(2n-1)\pi}{2n} \right].$$

(c) Show that if one approximates

$$\int_0^\pi \frac{\sin x}{x} dx$$

using the partition $x_k := (2k-1)(\pi/2n)$, $k = 1, 2, \dots, n$, $\Delta x_k = \pi/n$ and choosing the midpoint of

each interval as the place to evaluate the integrand, then

$$\begin{aligned} \int_0^\pi \frac{\sin x}{x} dx &\approx \frac{\sin(\pi/2n)}{\pi/2n} \frac{\pi}{n} + \cdots \\ &\quad + \frac{\sin[(2n-1)\pi/2n]}{(2n-1)\pi/2n} \frac{\pi}{n} \\ &= \frac{\pi}{2} f_{2n-1}\left(\frac{\pi}{2n}\right). \end{aligned}$$

(d) Use the result of part (c) to show that the overshoot satisfies

$$\lim_{n \rightarrow \infty} f_{2n-1}\left(\frac{\pi}{2n}\right) = \frac{2}{\pi} \int_0^\pi \frac{\sin x}{x} dx.$$

(e) Using the result of part (d) and a numerical integration algorithm (such as Simpson's rule, Appendix C) for the **sine integral function**

$$Si(z) := \int_0^z \frac{\sin x}{x} dx,$$

show that $\lim_{n \rightarrow \infty} f_{2n-1}(\pi/(2n)) \approx 1.18$. Thus, the approximations overshoot the true value of $f(0^+) = 1$ by 0.18, or 9% of the jump from $f(0^-)$ to $f(0^+)$.

10.4 Fourier Cosine and Sine Series

A typical problem encountered in using separation of variables to solve a partial differential equation is the problem of representing a function defined on some finite interval by a trigonometric series consisting of only sine functions or only cosine functions. For example, in Section 10.2, equation (25), page 569, we needed to express the initial values $u(x, 0) = f(x)$, $0 < x < L$, of the solution to the initial-boundary value problem associated with the problem of a vibrating string as a trigonometric series of the form

$$(1) \quad f(x) = \sum_{n=1}^{\infty} a_n \sin\left(\frac{n\pi x}{L}\right).$$

Recalling that the Fourier series for an odd function defined on $[-L, L]$ consists entirely of sine terms, we might try to achieve (1) by artificially extending the function $f(x)$, $0 < x < L$, to the interval $(-L, L)$ in such a way that the extended function is odd. This is accomplished by defining the function

$$f_o(x) := \begin{cases} f(x), & 0 < x < L, \\ -f(-x), & -L < x < 0, \end{cases}$$

and extending $f_o(x)$ to all x using $2L$ -periodicity.[†] Since $f_o(x)$ is an odd function, it has a Fourier series consisting entirely of sine terms. Moreover, $f_o(x)$ is an extension of $f(x)$,

[†]Strictly speaking, we have extended $f_o(x)$ to all x other than the integer multiples of L . Continuity considerations often suggest appropriate values for the extended functions at some of these points, as well. Figure 10.13 on page 589 illustrates this.

since $f_o(x) = f(x)$ on $(0, L)$. This extension is called the **odd $2L$ -periodic extension** of $f(x)$. The resulting Fourier series expansion is called a half-range expansion for $f(x)$, since it represents the function $f(x)$ on $(0, L)$, which is half of the interval $(-L, L)$ where it represents $f_o(x)$.

In a similar fashion, we can define the **even $2L$ -periodic extension** of $f(x)$ as the function

$$f_e(x) := \begin{cases} f(x), & 0 < x < L, \\ f(-x), & -L < x < 0, \end{cases}$$

with $f_e(x + 2L) = f_e(x)$.

To illustrate the various extensions, let's consider the function $f(x) = x$, $0 < x < \pi$. If we extend $f(x)$ to the interval $(-\pi, \pi)$ using π -periodicity, then the extension $\tilde{f}$ is given by

$$\tilde{f}(x) = \begin{cases} x, & 0 < x < \pi, \\ x + \pi, & -\pi < x < 0, \end{cases}$$

with $\tilde{f}(x + 2\pi) = \tilde{f}(x)$. In Problem 14 of Exercises 10.3, the Fourier series for $\tilde{f}(x)$ was found to be

$$\tilde{f}(x) \sim \frac{\pi}{2} - \sum_{n=1}^{\infty} \frac{1}{n} \sin 2nx,$$

which consists of both odd functions (the sine terms) and even functions (the constant term), since the π -periodic extension $\tilde{f}(x)$ is neither an even nor an odd function. The odd 2π -periodic extension of $f(x)$ is just $f_o(x) = x$, $-\pi < x < \pi$, which has the Fourier series expansion

$$(2) \quad f_o(x) \sim 2 \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \sin nx$$

(see Problem 9 in Exercises 10.3). Because $f_o(x) = f(x)$ on the interval $(0, \pi)$, the expansion in (2) is a half-range expansion for $f(x)$. The even 2π -periodic extension of $f(x)$ is the function $f_e(x) = |x|$, $-\pi < x < \pi$, which has the Fourier series expansion

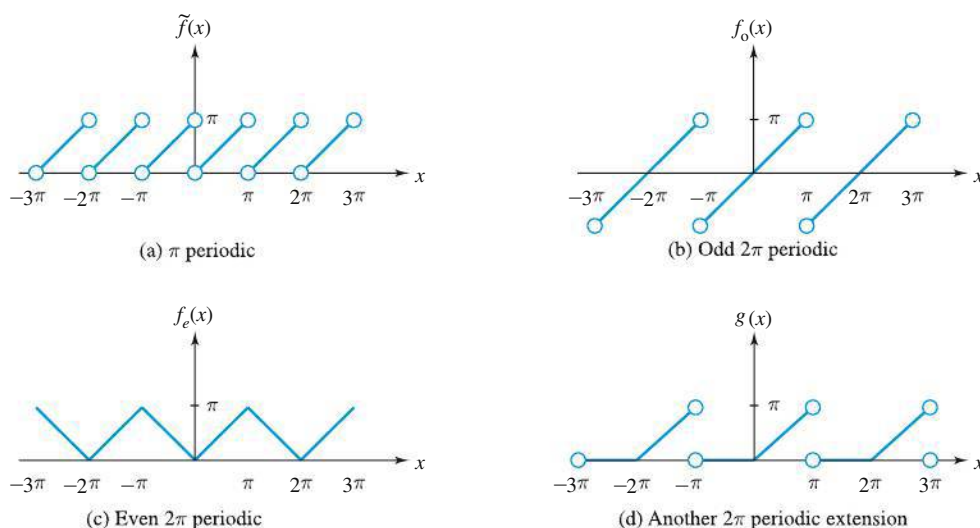
$$(3) \quad f_e(x) = \frac{\pi}{2} - \frac{4}{\pi} \sum_{n=0}^{\infty} \frac{1}{(2n+1)^2} \cos(2n+1)x$$

(see Problem 10 in Exercises 10.3).

The preceding three extensions, the π -periodic function $\tilde{f}(x)$, the odd 2π -periodic function $f_o(x)$, and the even 2π -periodic function $f_e(x)$, are natural extensions of $f(x)$. There are many other ways of extending $f(x)$. For example, the function

$$g(x) = \begin{cases} x, & 0 < x < \pi, \\ 0, & -\pi < x < 0, \end{cases} \quad g(x + 2\pi) = g(x),$$

which we studied in Example 3 of Section 10.3, is also an extension of $f(x)$. However, its Fourier series contains both sine and cosine terms and hence is not as useful as previous extensions. The graphs of these extensions of $f(x)$ are given in Figure 10.13 on page 589.

Figure 10.13 Extensions of $f(x) = x, 0 < x < \pi$

The Fourier series expansions for $f_o(x)$ and $f_e(x)$ given in (2) and (3) represent $f(x)$ on the interval $(0, \pi)$ (actually, they equal $f(x)$ on $(0, \pi)$). This motivates the following definitions.

Fourier Cosine and Sine Series

Definition 2. Let $f(x)$ be piecewise continuous on the interval $[0, L]$. The **Fourier cosine series** of $f(x)$ on $[0, L]$ is

$$(4) \quad \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{L},$$

where

$$(5) \quad a_n = \frac{2}{L} \int_0^L f(x) \cos \frac{n\pi x}{L} dx, \quad n = 0, 1, \dots$$

The **Fourier sine series** of $f(x)$ on $[0, L]$ is

$$(6) \quad \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{L},$$

where

$$(7) \quad b_n = \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi x}{L} dx, \quad n = 1, 2, \dots$$

The trigonometric series in (4) is just the Fourier series for $f_e(x)$, the even $2L$ -periodic extension of $f(x)$, and that in (6) is the Fourier series for $f_o(x)$, the odd $2L$ -periodic extension of $f(x)$. These are called **half-range expansions** for $f(x)$.

Example 1 Compute the Fourier sine series for

$$f(x) = \begin{cases} x, & 0 \leq x \leq \pi/2, \\ \pi - x, & \pi/2 \leq x \leq \pi. \end{cases}$$

Solution Using formula (7) with $L = \pi$, we find

$$\begin{aligned} b_n &= \frac{2}{\pi} \int_0^\pi f(x) \sin nx \, dx = \frac{2}{\pi} \int_0^{\pi/2} x \sin nx \, dx + \frac{2}{\pi} \int_{\pi/2}^\pi (\pi - x) \sin nx \, dx \\ &= \frac{2}{\pi n^2} \int_0^{\pi n/2} u \sin u \, du + 2 \int_{\pi/2}^\pi \sin nx \, dx - \frac{2}{\pi n^2} \int_{\pi n/2}^{\pi n} u \sin u \, du \\ &= \frac{2}{\pi n^2} [\sin u - u \cos u] \Big|_0^{\pi n/2} - \frac{2}{n} \left[\cos \pi n - \cos \frac{n\pi}{2} \right] \\ &\quad - \frac{2}{\pi n^2} [\sin u - u \cos u] \Big|_{\pi n/2}^{\pi n} \\ &= \frac{4}{\pi n^2} \sin \frac{n\pi}{2} = \begin{cases} 0, & n \text{ even}, \\ \frac{4(-1)^{(n-1)/2}}{\pi n^2}, & n \text{ odd}. \end{cases} \end{aligned}$$

So on letting $n = 2k + 1$, we find the Fourier sine series for $f(x)$ to be

$$(8) \quad \frac{4}{\pi} \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)^2} \sin(2k+1)x = \frac{4}{\pi} \left\{ \sin x - \frac{1}{9} \sin 3x + \frac{1}{25} \sin 5x + \cdots \right\}. \quad \blacklozenge$$

Since, in Example 1, the function $f(x)$ is continuous and $f'(x)$ is piecewise continuous on $(0, \pi)$, it follows from Theorem 2 on pointwise convergence of Fourier series that

$$f(x) = \frac{4}{\pi} \left\{ \sin x - \frac{1}{9} \sin 3x + \frac{1}{25} \sin 5x - \frac{1}{49} \sin 7x + \cdots \right\}$$

for all x in $[0, \pi]$.

Let's return to the problem of heat flow in one dimension.

Example 2 Find the solution to the heat flow problem

$$(9) \quad \frac{\partial u}{\partial t} = 2 \frac{\partial^2 u}{\partial x^2}, \quad 0 < x < \pi, \quad t > 0,$$

$$(10) \quad u(0, t) = u(\pi, t) = 0, \quad t > 0,$$

$$(11) \quad u(x, 0) = \begin{cases} x, & 0 < x \leq \pi/2, \\ \pi - x, & \pi/2 \leq x < \pi. \end{cases}$$

Solution Comparing equation (9) with equation (1) in Section 10.2 (page 563), we see that $\beta = 2$ and $L = \pi$. Hence, we need only represent $u(x, 0) = f(x)$ in a Fourier sine series of the form

$$\sum_{n=1}^{\infty} c_n \sin nx.$$

In Example 1 we obtained this expansion and showed that

$$c_n = \frac{4}{\pi n^2} \sin \frac{n\pi}{2} = \begin{cases} 0, & n \text{ even}, \\ \frac{4(-1)^{(n-1)/2}}{\pi n^2}, & n \text{ odd}. \end{cases}$$

Hence, from equation (15) on page 566, the solution to the heat flow problem (9)–(11) is

$$\begin{aligned} u(x, t) &= \sum_{n=1}^{\infty} c_n e^{-2n^2 t} \sin nx \\ &= \frac{4}{\pi} \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)^2} e^{-2(2k+1)^2 t} \sin(2k+1)x \\ &= \frac{4}{\pi} \left\{ e^{-2t} \sin x - \frac{1}{9} e^{-18t} \sin 3x + \frac{1}{25} e^{-50t} \sin 5x + \cdots \right\}. \end{aligned}$$

A sketch of a partial sum for $u(x, t)$ is displayed in Figure 10.14. ♦

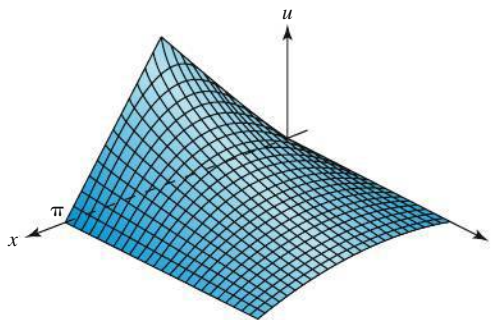


Figure 10.14 Partial sum for $u(x, t)$ in Example 2

10.4 EXERCISES

In Problems 1–4, determine (a) the π -periodic extension $\tilde{f}$, (b) the odd 2π -periodic extension f_o , and (c) the even 2π -periodic extension f_e for the given function f and sketch their graphs.

1. $f(x) = x^2$, $0 < x < \pi$
2. $f(x) = \sin 2x$, $0 < x < \pi$
3. $f(x) = \begin{cases} 0, & 0 < x < \pi/2, \\ 1, & \pi/2 < x < \pi \end{cases}$
4. $f(x) = \pi - x$, $0 < x < \pi$

In Problems 5–10, compute the Fourier sine series for the given function.

5. $f(x) = -1$, $0 < x < 1$
6. $f(x) = \cos x$, $0 < x < \pi$

7. $f(x) = x^2$, $0 < x < \pi$
8. $f(x) = \pi - x$, $0 < x < \pi$
9. $f(x) = x - x^2$, $0 < x < 1$
10. $f(x) = e^x$, $0 < x < 1$

In Problems 11–16, compute the Fourier cosine series for the given function.

11. $f(x) = \pi - x$, $0 < x < \pi$
12. $f(x) = 1 + x$, $0 < x < \pi$
13. $f(x) = e^x$, $0 < x < 1$
14. $f(x) = e^{-x}$, $0 < x < 1$
15. $f(x) = \sin x$, $0 < x < \pi$
16. $f(x) = x - x^2$, $0 < x < 1$